

A strictness criterion for higher-order asymptotic models

Partial solution packet for Question 5.18 of arXiv:1806.08749

11 August 2026

Substantial partial answer. For every positive integer m and every countable ordinal $\eta \geq m + 2$, there is a reflexive Banach space X such that

$$\text{AM}_m(X) \neq \text{AM}_\eta(X).$$

More generally, a spreading basis in $\text{SM}_\eta(X) \setminus \text{SM}_{\xi+1}(X)$ witnesses $\text{AM}_\xi(X) \neq \text{AM}_\eta(X)$. The adjacent case $\eta = \xi + 1$ and pairs with infinite lower level ξ are not settled here.

Source question

Calderon-Garcia and Garcia-Ferreira define the classes $\text{AM}_\xi(X)$ of ξ -asymptotic models and prove their monotonicity. They ask whether every pair of distinct countable ordinals can be separated by a Banach space [1].

weakly generated asymptotic models $A_1, A_2, \dots, A_n$ starting with A has a an ending space either l_p for $p \in [1, \infty)$ or c_0 .

We finish the paper by listing some questions

As in the spreading case we also have that $\text{AM}_\xi(X) \subseteq \text{AM}_\eta(X)$ provided that $\xi < \eta < \omega_1$.

Question 5.18. For distinct $\xi < \eta < \omega_1$ does there exist a Banach space X such that

$$\text{AM}_\xi(X) \neq \text{AM}_\eta(X) ?$$

For a $\mathcal{F} \subseteq \text{FIN}$ and $t \in \text{FIN}$ we denote by $\mathcal{F}_t = \{s \in \text{FIN} : t < s \text{ and } t \frown s \in \mathcal{F}\}$. If $B \subseteq \text{FIN}$ is a uniform barrier, then it is known that for every $t \in \text{FIN}$ such that $\mathcal{B}_t$ has uniformity greater than 1 it is possible to find $s \in \text{FIN}$ such that $\mathcal{B}_{t \frown s}$ has uniformity exactly one.

The key input from the source is Theorem 5.10:

- (i) $\text{SM}_\xi(X) \subseteq \text{AM}_\xi(X)$;
- (ii) if $(f_i) \in \text{AM}_\xi(X)$ and a subsequence of (f_i) generates the spreading model (e_i) , then $(e_i) \in \text{SM}_{\xi+1}(X)$.

Here $\text{SM}_\xi(X)$ denotes the class of ξ -spreading models. The second statement provides a necessary condition for a spreading basis to occur already at asymptotic level ξ .

The one-order obstruction

Proposition 1. Let $0 < \xi < \omega_1$, let X be a Banach space, and let (e_i) be a normalized Schauder basic spreading sequence. If

$$(e_i) \in \text{AM}_\xi(X),$$

then $(e_i) \in \text{SM}_{\xi+1}(X)$.

Proof. A spreading sequence generates itself, isometrically, as an ordinary spreading model: for every increasing $n_1 < \dots < n_k$ and scalars $(a_i)_{i=1}^k$,

$$\left\| \sum_{i=1}^k a_i e_{n_i} \right\| = \left\| \sum_{i=1}^k a_i e_i \right\|.$$

Apply part (ii) of Theorem 5.10 of the source to $(f_i) = (e_i)$ and to any subsequence. This puts (e_i) in $\text{SM}_{\xi+1}(X)$. $\square$

Corollary 1 (strict-spreading witness). *Let $0 < \xi < \eta < \omega_1$. If X admits a normalized Schauder basic spreading sequence (e_i) such that*

$$(e_i) \in \text{SM}_{\eta}(X) \setminus \text{SM}_{\xi+1}(X),$$

then $\text{AM}_{\xi}(X) \neq \text{AM}_{\eta}(X)$.

Proof. By Theorem 5.10(i), $(e_i) \in \text{AM}_{\eta}(X)$. Proposition 1 shows that (e_i) cannot belong to $\text{AM}_{\xi}(X)$. $\square$

This criterion isolates the exact one-level buffer in the present method. It turns any sufficiently separated strictness result for higher-order spreading models into strictness of the asymptotic-model hierarchy.

Unconditional finite-level separation

Argyros, Kanellopoulos, and Tyros construct, for every positive integer k , a reflexive Banach space $\mathfrak{X}_{k+1}$ with an unconditional basis such that

$$(\ell_1 \text{ basis}) \in \text{SM}_{k+1}(\mathfrak{X}_{k+1}) \quad \text{but} \quad (\ell_1 \text{ basis}) \notin \text{SM}_k(\mathfrak{X}_{k+1}). \quad (1)$$

The first membership is isometric; the second excludes even equivalence to the ℓ_1 basis [2, Section 12].

Theorem 1. *For every positive integer m and every countable ordinal η satisfying $\eta \geq m + 2$, there is a reflexive Banach space X for which*

$$\text{AM}_m(X) \subsetneq \text{AM}_{\eta}(X).$$

In particular, Question 5.18 has an affirmative answer for every pair of positive finite levels $m < \eta$ with $\eta - m \geq 2$.

Proof. Take $X = \mathfrak{X}_{m+2}$ in (1), so the ℓ_1 basis belongs to $\text{SM}_{m+2}(X)$ but not to $\text{SM}_{m+1}(X)$. The source's inclusions give

$$\text{SM}_{m+2}(X) \subseteq \text{AM}_{m+2}(X) \subseteq \text{AM}_{\eta}(X),$$

so the ℓ_1 basis lies in $\text{AM}_{\eta}(X)$. If it belonged to $\text{AM}_m(X)$, Proposition 1 would put it in $\text{SM}_{m+1}(X)$, contradicting (1). Finally $\text{AM}_m(X) \subseteq \text{AM}_{\eta}(X)$ by monotonicity, so the inclusion is proper. $\square$

Deep upgrade attempts and the adjacent obstruction

Three further routes were checked before stopping at the partial result.

1. Peeling one coordinate from the finite-order example. The tempting adjacent-level construction is to turn the canonical $(k+2)$ -sequence of $\mathfrak{X}_{k+2}$ into a $(k+1) \times \mathbb{N}$ matrix by inserting the row number as an extra coordinate. This does not work. The norm of $\mathfrak{X}_{k+2}$ detects ℓ_1 on *Schreier plegmatic* families; if $F_1 < \dots < F_{k+2}$ are the coordinate blocks, their common cardinality must satisfy

$$|F_1| \leq \min F_1.$$

Putting the row number in the first coordinate fixes the least first coordinate while the tested combination length tends to infinity, so the resulting families eventually fail the Schreier condition. Inserting the coordinate between later columns also fails uniformly: a matrix entry is chosen row by row, whereas the new coordinate would have to lie simultaneously above the largest coordinate in one whole plegma column and below the smallest coordinate in the next. Those extrema depend on the entire tested family.

2. Using separation for a distinguished subset. The higher-order spreading-model paper constructs, for a regular thin family $\mathcal{G}$ of order η , a space $X_{\mathcal{G}}$ whose canonical set $A = \{e_s : s \in \mathcal{G}\}$ has a model at level η absent from all lower spreading classes generated *inside* A [3]. This does not feed Corollary 1: $\text{SM}_{\zeta}(X_{\mathcal{G}})$ permits arbitrary vectors of the completed space, not just elements of A . Excluding lower-level models of arbitrary block combinations is precisely the missing step.

3. Transfinite strictness announcements. The higher-order spreading-model literature announces whole-space strictness examples at countable limit levels, but the source inspected here labels them as forthcoming rather than proving them [3, Remark 3]. They were therefore not used as a theorem. If a proved witness $(e_i) \in \text{SM}_{\eta}(X) \setminus \bigcup_{\zeta < \eta} \text{SM}_{\zeta}(X)$ is supplied, Corollary 1 immediately separates $\text{AM}_{\xi}(X)$ from $\text{AM}_{\eta}(X)$ for every $\xi + 1 < \eta$.

Thus the method is complete for the range stated in Theorem 1, but it genuinely loses one order. New geometry, rather than a reindexing of the known basis, appears necessary for adjacent successor levels.

Bounded novelty check and scope

On 11 August 2026, the exact arXiv id and Question 5.18 text, together with searches for the AM_{ξ} hierarchy, plegma asymptotic models, and strict higher-order spreading models, were checked in the run indexes and by bounded web/arXiv search. The searches returned the source, the finite- and higher-order spreading-model papers, and later work on related asymptotic structures, but no statement of Proposition 1 as a hierarchy criterion or of Theorem 1. This is evidence for novelty, not a definitive guarantee.

The packet is a partial answer to Question 5.18. It does not settle adjacent pairs $\eta = \xi + 1$, nor pairs whose lower level ξ is infinite.

References

- [1] E. A. Calderon-Garcia and S. Garcia-Ferreira, *Asymptotic models via plegma families*, arXiv:1806.08749, 2018.
- [2] S. A. Argyros, V. Kanellopoulos, and K. Tyros, *Finite order spreading models*, Adv. Math. **234** (2013), 574–617; arXiv:1105.2732.
- [3] S. A. Argyros, V. Kanellopoulos, and K. Tyros, *Higher order spreading models*, Fund. Math. **221** (2013), 23–68; arXiv:1202.6390.